

Lizhong Hu

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Profile

Software Engineering undergraduate in a Sino-American dual-degree program jointly operated by Quanzhou University of Information Engineering and Slippery Rock University. Interested in algorithms, software systems, information retrieval, and reliable intelligent systems, with experience in competitive programming, back-end development, Linux/Docker infrastructure, and live sports broadcast technical operations.

Research Interests

Intelligent Software Engineering; Programming Support and Online Judge Systems; Algorithms and Data Structures; Information Retrieval and Retrieval-Augmented Generation.

Education

Quanzhou University of Information Engineering 2023–2027 (*expected*)
Software Engineering (Sino-foreign Cooperation) GPA: 3.10/4.00

Slippery Rock University
Information Systems, Bachelor of Science (BS) GPA: 3.583/4.000
Four-year Sino-American 4+0 dual-degree undergraduate program conducted in China and officially recognized by the Ministry of Education of China.

Selected Projects

Personal Knowledge Question-Answering System with RAG *Ongoing*
Undergraduate thesis

- Developing a FastAPI-based personal knowledge question-answering system using retrieval-augmented generation, with document retrieval, knowledge grounding, and LLM-based responses.

DOMjudge / Xboard / 3x-ui / Cloudflare Infrastructure *Sep 2024–Present*

- Maintain a containerized DOMjudge environment for university training and contests; administer Xboard and 3x-ui for remote access, with Cloudflare DNS/CDN and Linux/Docker infrastructure.

ETi+ Back-End API Development *Mar 2025–Apr 2025*

- Built authentication, data aggregation, and testing APIs with Flask, PostgreSQL, and MongoDB; automated pre-deployment regression checks with Postman.

Technical Experience

Sinotech Xinrui (Beijing) Co., Ltd. *7–22 Sep 2026*
TOC Operator / Gifu, Japan

Project: 20th Asian Games Aichi-Nagoya 2026 — broadcast technical support.
Football, Gifu Nagaragawa Stadium.

Host broadcaster: China Media Group-International Media Port (CMG-IMP).

- Checked equipment power, intercom quality, and primary/backup broadcast feeds before matches in coordination with the broadcast team.
- Monitored TOC signals during matches and provided on-site technical support throughout a 16-day assignment.
- Helped engineers isolate an optical-transmitter fault through receiver replacement and attenuation tests; the monitor feed was restored after post-match replacement and end-to-end verification.

NOI-Pre Problem Bank — Matiji *Jul 2024–Oct 2024*
Problem Tester

In collaboration with China Computer Federation (CCF).

- Tested NOI-Pre programming problems; created test cases, checked solution correctness through targeted submissions, and reported issues with test data and judge results.

Competitive Programming & Awards

- 2026 National Invitational of CCPC (Fujian), The 13th Fujian Collegiate Programming Contest** — First accepted solution to Problem A.
- The 2024 ICPC Asia Kunming Regional Contest** — Onsite Contestant, 2024.
- 15th Lanqiao Cup** — First Prize, Fujian Provincial Round, Python Programming (University Group B), 2024.

Technical Skills

Languages & Algorithms: C++, Python, Java, JavaScript, Go; algorithms and data structures, competitive programming.

Back-End & Data: FastAPI, Flask, RuoYi, PostgreSQL, MongoDB, MySQL, REST API design and integration.

Systems & DevOps: Linux, Docker, Git, SQL tooling, DOMjudge deployment, Xboard and 3x-ui administration, Cloudflare DNS/CDN.

Information Retrieval & AI Systems: Retrieval-Augmented Generation (RAG), LLM API integration, prompt design.